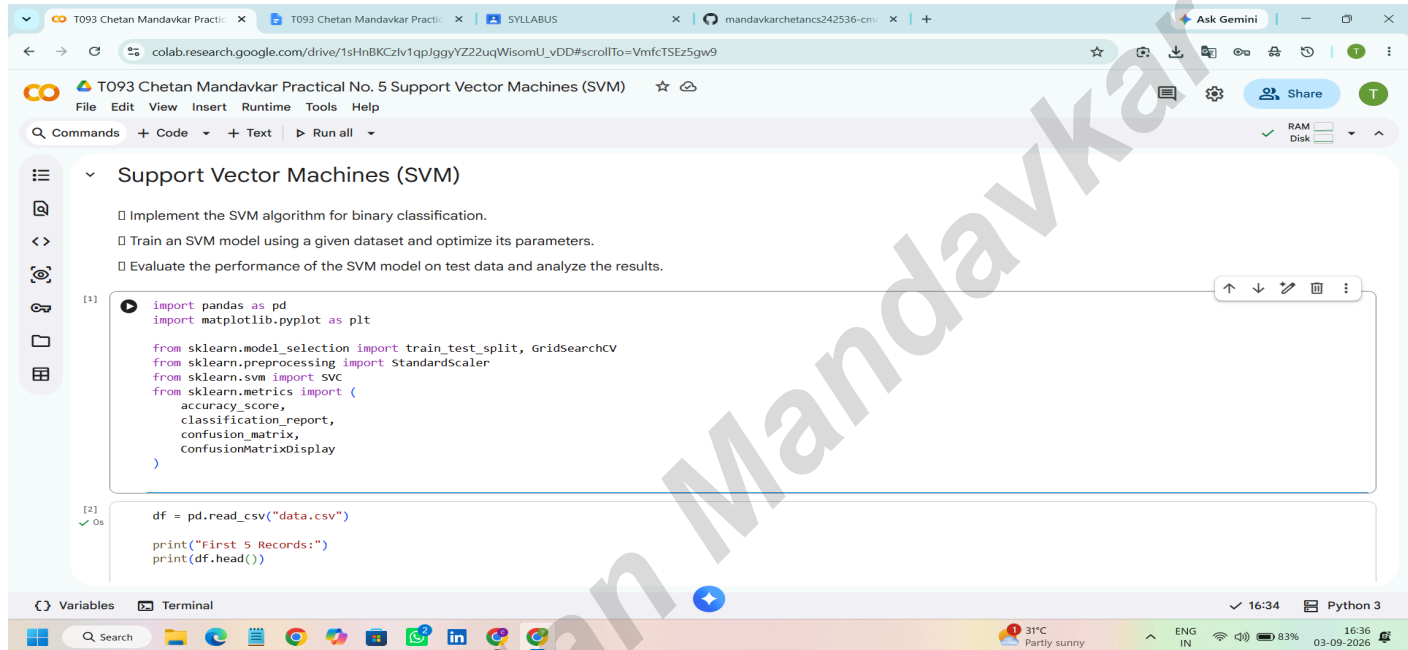


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Aim:- Support Vector Machines (SVM)

Implement the SVM algorithm for binary classification. Train an SVM model using a given dataset and optimize its parameters. Evaluate the performance of the SVM model on test data and analyze the results.



The screenshot shows a Google Colab notebook with the title "Support Vector Machines (SVM)". The notebook has a menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu bar is a toolbar with "Commands", "Code", "Text", and "Run all". The notebook content includes a list of tasks: "Implement the SVM algorithm for binary classification.", "Train an SVM model using a given dataset and optimize its parameters.", and "Evaluate the performance of the SVM model on test data and analyze the results." The code cell [1] contains the following Python code:

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, GridSearchcv
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix,
    confusionMatrixDisplay
)
```

The code cell [2] contains the following Python code:

```
df = pd.read_csv("data.csv")

print("First 5 Records:")
print(df.head())
```

The notebook interface also shows a "Variables" panel on the left and a "Terminal" panel at the bottom. The system tray at the bottom indicates the time is 16:34 and the date is 03-09-2026.



The screenshot shows the same Google Colab notebook, but now the code cell [2] has been executed, and the output is displayed. The output shows the first 5 records of the dataset and the dataset shape.

```
First 5 Records:
   id diagnosis  radius_mean  texture_mean  perimeter_mean  area_mean \
0   842302     M         10.38         17.99         132.80        1801.0
1   842517     M         20.57         17.77         132.90        1326.0
2   84300903    M         19.69         21.25         130.00        1203.0
3   84348301    M         11.42         20.38         77.58         386.1
4   84358402    M         20.29         14.34         135.10        1297.0

   smoothness_mean  compactness_mean  concavity_mean  concave points_mean \
0         0.11840         0.27760         0.3001         0.14710
1         0.08474         0.07864         0.0869         0.07017
2         0.10960         0.15990         0.1974         0.12790
3         0.14250         0.28390         0.2414         0.10520
4         0.10030         0.13280         0.1980         0.10430

   texture_worst  perimeter_worst  area_worst  smoothness_worst \
0         17.33         184.60        2019.0         0.1622
1         23.41         158.80        1956.0         0.1238
2         25.53         152.50        1709.0         0.1444
3         26.50         98.87         567.7         0.2098
4         16.67         152.20        1575.0         0.1374

   compactness_worst  concavity_worst  concave points_worst  symmetry_worst \
0         0.11840         0.27760         0.3001         0.14710
1         0.08474         0.07864         0.0869         0.07017
2         0.10960         0.15990         0.1974         0.12790
3         0.14250         0.28390         0.2414         0.10520
4         0.10030         0.13280         0.1980         0.10430
```

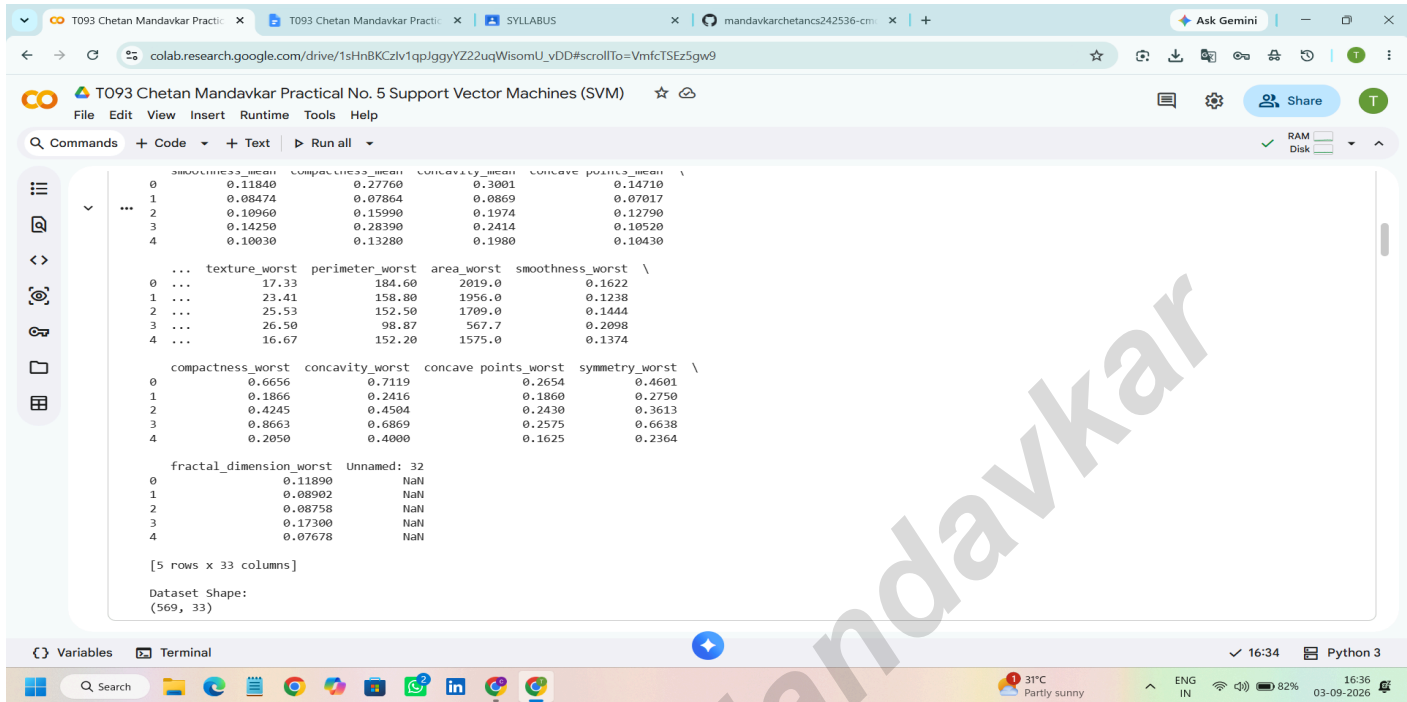
The notebook interface also shows a "Variables" panel on the left and a "Terminal" panel at the bottom. The system tray at the bottom indicates the time is 16:34 and the date is 03-09-2026.

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```
compactness_worst compactness_mean compactness_worst compactness_mean \
0 0.11840 0.27760 0.3001 0.14710
1 0.08474 0.07864 0.0869 0.07017
2 0.10960 0.15990 0.1974 0.12790
3 0.14250 0.28390 0.2414 0.10520
4 0.10030 0.13280 0.1980 0.10430

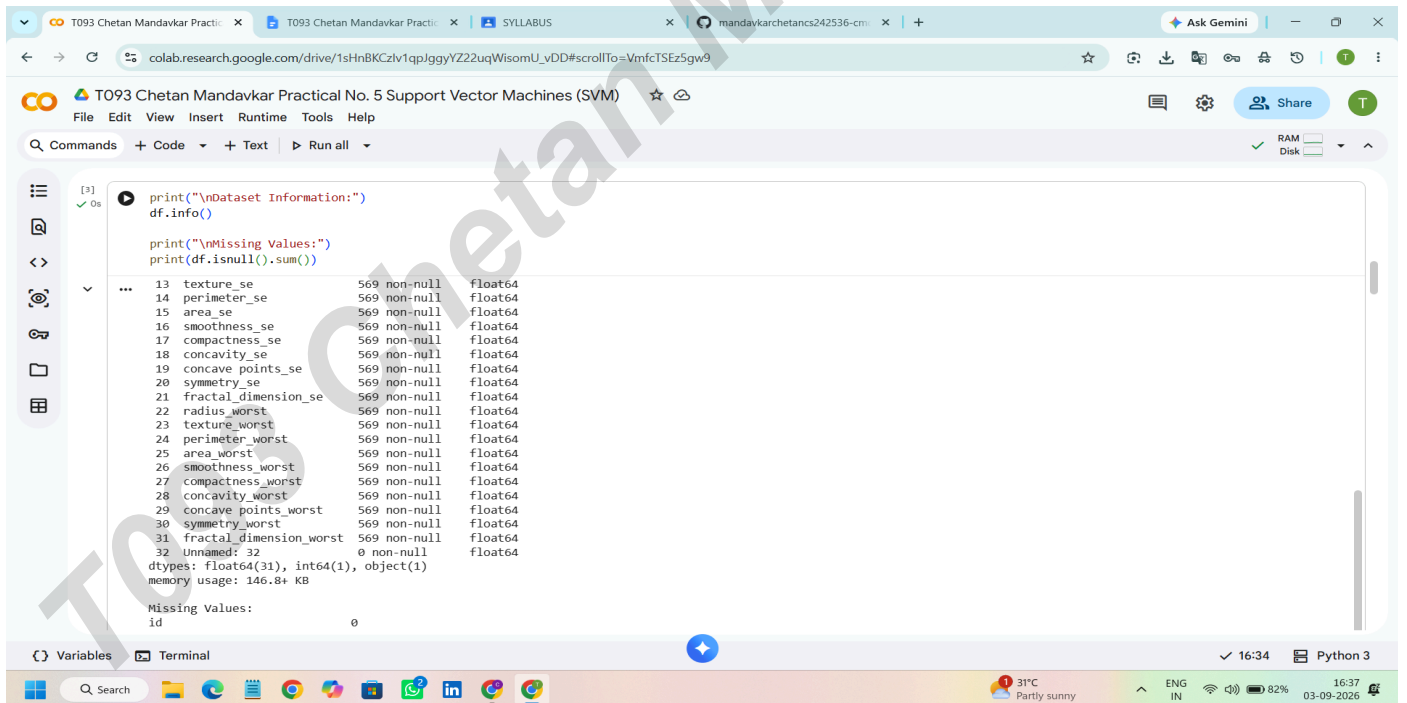
... texture_worst perimeter_worst area_worst smoothness_worst \
0 17.33 184.60 2019.0 0.1622
1 23.41 158.80 1956.0 0.1238
2 25.53 152.50 1709.0 0.1444
3 26.50 98.87 567.7 0.2098
4 16.67 152.20 1575.0 0.1374

compactness_worst concavity_worst concave points_worst symmetry_worst \
0 0.6656 0.7119 0.2654 0.4601
1 0.1866 0.2416 0.1860 0.2750
2 0.4245 0.4504 0.2430 0.3613
3 0.8663 0.6869 0.2575 0.6638
4 0.2050 0.4000 0.1625 0.2364

fractal_dimension_worst Unnamed: 32
0 0.11890 NaN
1 0.08902 NaN
2 0.08758 NaN
3 0.17300 NaN
4 0.07678 NaN

[5 rows x 33 columns]

Dataset Shape:
(569, 33)
```



```
[3] print("\nDataset Information:")
df.info()

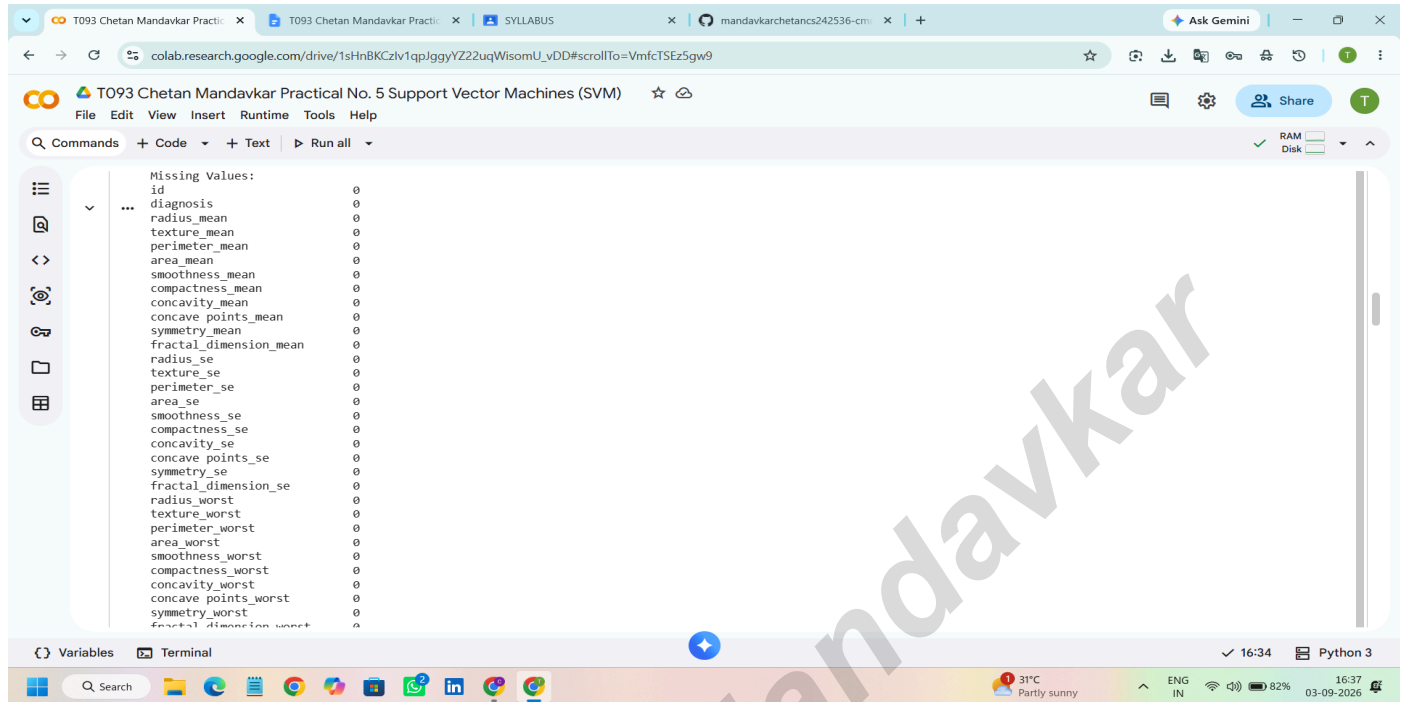
print("\nMissing Values:")
print(df.isnull().sum())

...
13 texture_se 569 non-null float64
14 perimeter_se 569 non-null float64
15 area_se 569 non-null float64
16 smoothness_se 569 non-null float64
17 compactness_se 569 non-null float64
18 concavity_se 569 non-null float64
19 concave points_se 569 non-null float64
20 symmetry_se 569 non-null float64
21 fractal_dimension_se 569 non-null float64
22 radius_worst 569 non-null float64
23 texture_worst 569 non-null float64
24 perimeter_worst 569 non-null float64
25 area_worst 569 non-null float64
26 smoothness_worst 569 non-null float64
27 compactness_worst 569 non-null float64
28 concavity_worst 569 non-null float64
29 concave points_worst 569 non-null float64
30 symmetry_worst 569 non-null float64
31 fractal_dimension_worst 569 non-null float64
32 Unnamed: 32 0 non-null float64
dtypes: float64(31), int64(1), object(1)
memory usage: 146.8+ KB

Missing Values:
id 0
```

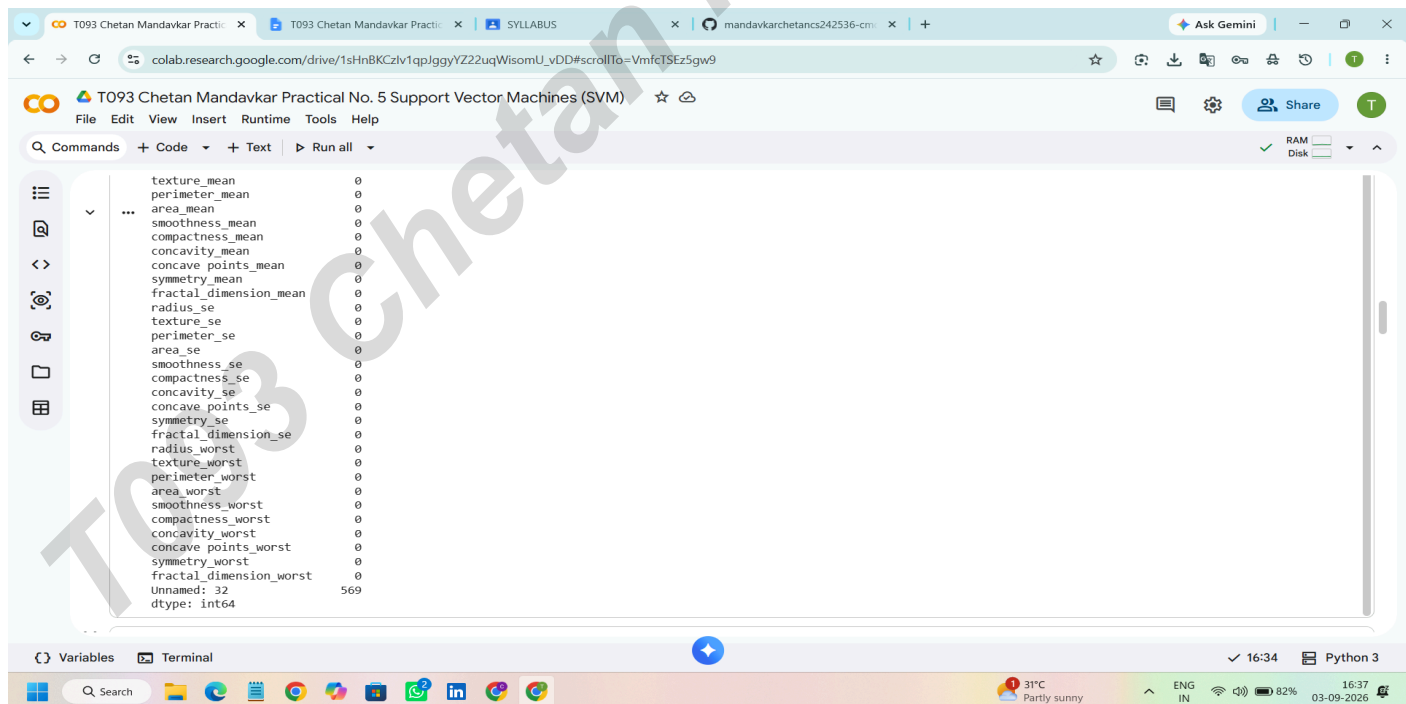
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The screenshot shows a Google Colab environment with a notebook titled "T093 Chetan Mandavkar Practical No. 5 Support Vector Machines (SVM)". The code cell displays a list of missing values for various features in a dataset. The features and their corresponding missing values are:

Feature	Missing Values
id	0
diagnosis	0
radius_mean	0
texture_mean	0
perimeter_mean	0
area_mean	0
smoothness_mean	0
compactness_mean	0
concavity_mean	0
concave points_mean	0
symmetry_mean	0
fractal_dimension_mean	0
radius_se	0
texture_se	0
perimeter_se	0
area_se	0
smoothness_se	0
compactness_se	0
concavity_se	0
concave points_se	0
symmetry_se	0
fractal_dimension_se	0
radius_worst	0
texture_worst	0
perimeter_worst	0
area_worst	0
smoothness_worst	0
compactness_worst	0
concavity_worst	0
concave points_worst	0
symmetry_worst	0
fractal_dimension_worst	0

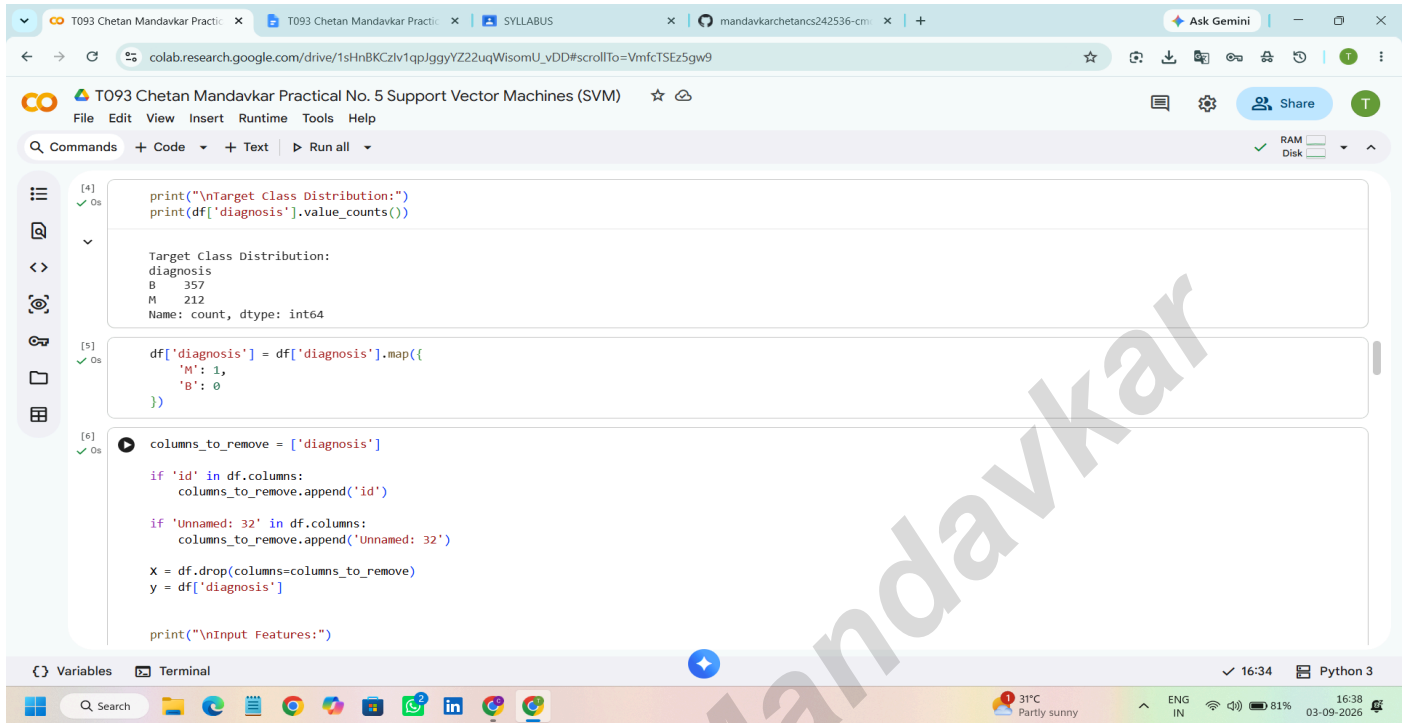


The screenshot shows the same Google Colab environment. The code cell displays the distribution of features in the dataset. The features and their corresponding distributions are:

Feature	Distribution
texture_mean	0
perimeter_mean	0
area_mean	0
smoothness_mean	0
compactness_mean	0
concavity_mean	0
concave points_mean	0
symmetry_mean	0
fractal_dimension_mean	0
radius_se	0
texture_se	0
perimeter_se	0
area_se	0
smoothness_se	0
compactness_se	0
concavity_se	0
concave points_se	0
symmetry_se	0
fractal_dimension_se	0
radius_worst	0
texture_worst	0
perimeter_worst	0
area_worst	0
smoothness_worst	0
compactness_worst	0
concavity_worst	0
concave points_worst	0
symmetry_worst	0
fractal_dimension_worst	0
Unnamed: 32	569
dtype:	int64

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Roll No. T093
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```
[4] print("\nTarget Class Distribution:")
    print(df['diagnosis'].value_counts())

Target Class Distribution:
diagnosis
B    357
M    212
Name: count, dtype: int64

[5] df['diagnosis'] = df['diagnosis'].map({
    'M': 1,
    'B': 0
})

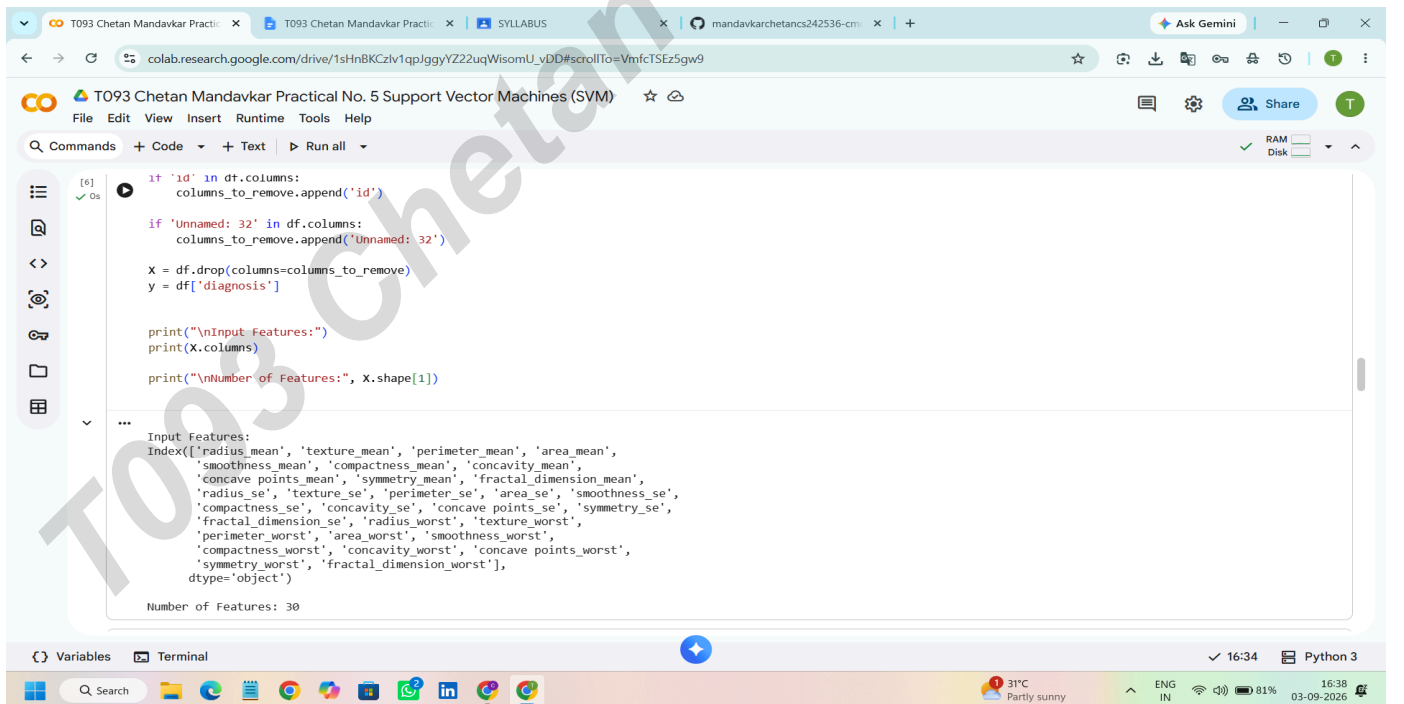
[6] columns_to_remove = ['diagnosis']

if 'id' in df.columns:
    columns_to_remove.append('id')

if 'Unnamed: 32' in df.columns:
    columns_to_remove.append('Unnamed: 32')

X = df.drop(columns=columns_to_remove)
y = df['diagnosis']

print("\nInput Features:")
```



```
[6] if 'id' in df.columns:
    columns_to_remove.append('id')

    if 'Unnamed: 32' in df.columns:
        columns_to_remove.append('Unnamed: 32')

    X = df.drop(columns=columns_to_remove)
    y = df['diagnosis']

    print("\nInput Features:")
    print(X.columns)

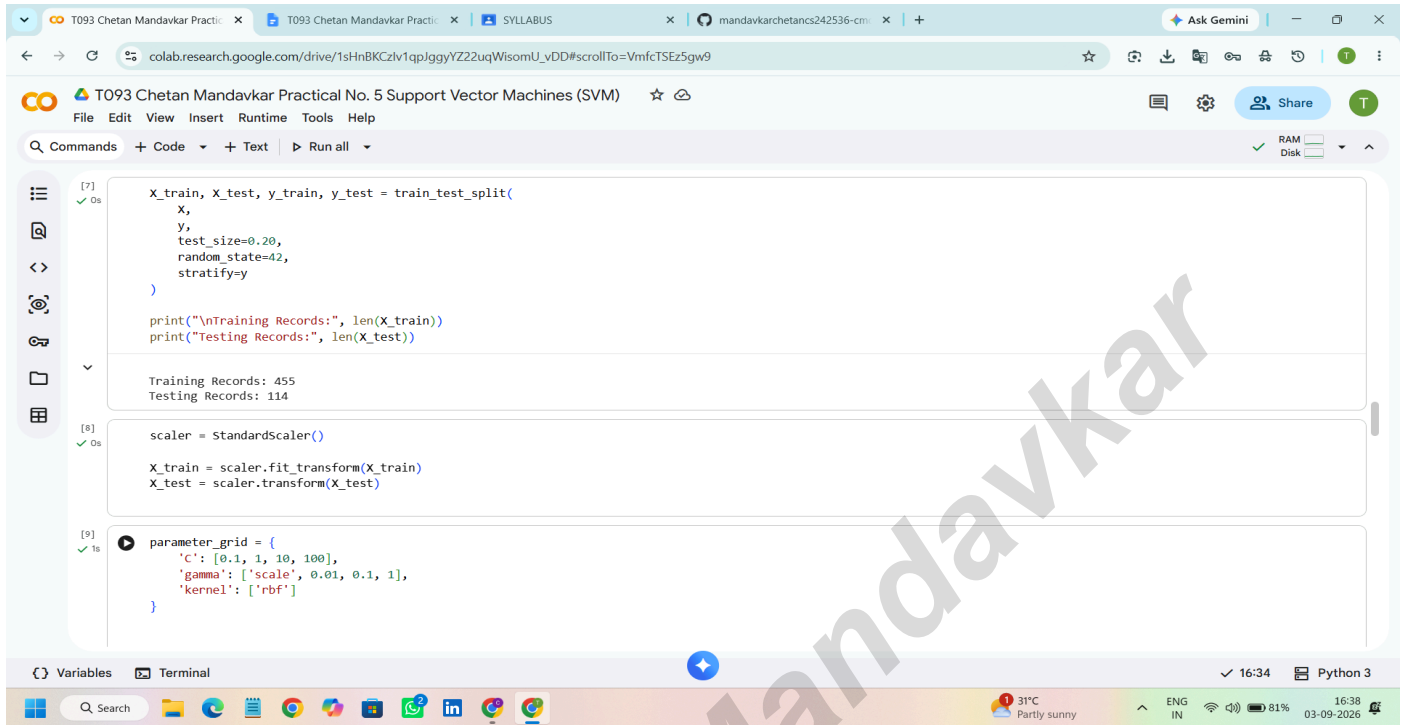
    print("\nNumber of Features:", X.shape[1])

...
Input Features:
Index(['radius_mean', 'texture_mean', 'perimeter_mean', 'area_mean',
      'smoothness_mean', 'compactness_mean', 'concavity_mean',
      'concave points_mean', 'symmetry_mean', 'fractal_dimension_mean',
      'radius_se', 'texture_se', 'perimeter_se', 'area_se', 'smoothness_se',
      'compactness_se', 'concavity_se', 'concave points_se', 'symmetry_se',
      'fractal_dimension_se', 'radius_worst', 'texture_worst',
      'perimeter_worst', 'area_worst', 'smoothness_worst',
      'compactness_worst', 'concavity_worst', 'concave points_worst',
      'symmetry_worst', 'fractal_dimension_worst'],
      dtype='object')

Number of Features: 30
```

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The screenshot shows a Google Colab notebook titled "T093 Chetan Mandavkar Practical No. 5 Support Vector Machines (SVM)". The code is written in Python and is being executed in a cell. The output shows the training and testing records for the SVM model.

```
[7] ✓ 0s
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

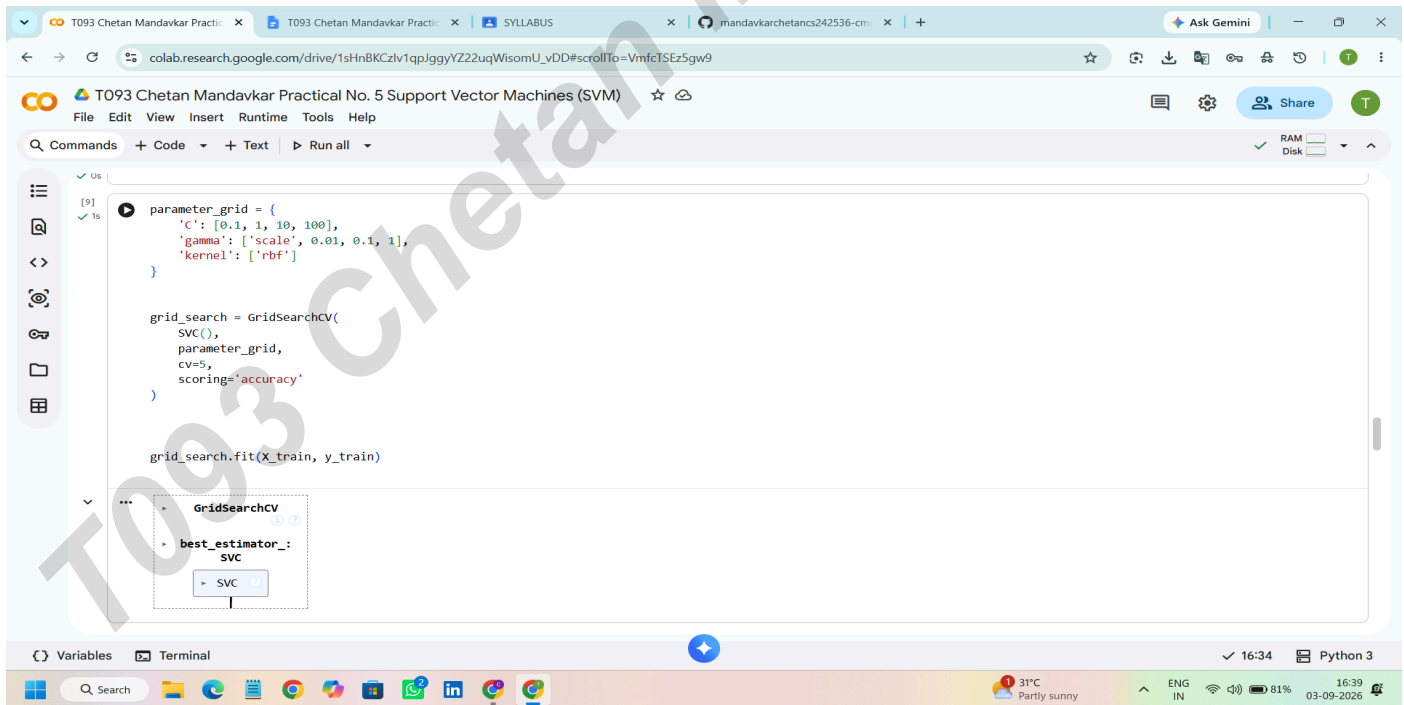
print("\nTraining Records:", len(X_train))
print("Testing Records:", len(X_test))

Training Records: 455
Testing Records: 114

[8] ✓ 0s
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

[9] ✓ 1s
parameter_grid = {
    'C': [0.1, 1, 10, 100],
    'gamma': ['scale', 0.01, 0.1, 1],
    'kernel': ['rbf']
}
```



The screenshot shows the same Google Colab notebook, but now the code is using GridSearchCV to find the best parameters for the SVM model. The output shows the best estimator found.

```
[9] ✓ 1s
parameter_grid = {
    'C': [0.1, 1, 10, 100],
    'gamma': ['scale', 0.01, 0.1, 1],
    'kernel': ['rbf']
}

grid_search = GridSearchCV(
    SVC(),
    parameter_grid,
    cv=5,
    scoring='accuracy'
)

grid_search.fit(X_train, y_train)

GridSearchCV
  best_estimator_:
    SVC
```

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Subject:- Artificial Intelligence


```
[10] ✓ Os
print("\nBest Parameters:")
print(grid_search.best_params_)

print("\nBest Cross-Validation Accuracy:")
print(grid_search.best_score_ * 100, "%")

Best Parameters:
{'C': 1, 'gamma': 'scale', 'kernel': 'rbf'}

Best Cross-Validation Accuracy:
97.58241758241759 %

[11] ✓ Os
svm_model = grid_search.best_estimator_

[12] ✓ Os
y_pred = svm_model.predict(X_test)

print("\nActual Values:")
print(y_test.values)

print("\nPredicted Values:")
print(y_pred)

...
Actual Values:
[0 1 0 1 0 0 1 0 0 0 1 0 1 0 0 0 1 0 0 0 0 0 1 1 1 0 0 1 1 1 1 0 1 1 0 0 0
 0 1 0 0 0 1 0 0 1 0 1 0 0 1 1 0 0 0 1 0 0 0 1 0 0 0 1 1 0 1 0 1 1 0 0
 0 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0]
```

colab.research.google.com/drive/1sHnKCzlv1qpLggyYZ22uqWisomU_vDD#scrollTo=VmfctSEz5gw9

T093 Chetan Mandavkar Practical No. 5 Support Vector Machines (SVM)

File Edit View Insert Runtime Tools Help

Commands Code Text Run all

RAM Disk

Best Parameters:
{'C': 1, 'gamma': 'scale', 'kernel': 'rbf'}

Best Cross-Validation Accuracy:
97.58241758241759 %

[11] ✓ 0s
svm_model = grid_search.best_estimator_

[12] ✓ 0s
y_pred = svm_model.predict(X_test)

print("\nActual Values:")
print(y_test.values)

print("\nPredicted Values:")
print(y_pred)

...
Actual Values:
[0 1 0 1 0 0 1 0 0 0 1 0 1 0 0 0 1 0 0 0 0 0 1 1 1 0 0 1 1 1 1 0 1 1 0 0 0
0 1 0 0 0 1 0 1 0 1 0 0 1 1 0 0 1 1 0 0 1 0 0 1 0 0 0 1 1 1 0 1 0 1 1 0
0 0 0 1 0 1 1 1 0 0 0 0 1 1 1 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 1 0
0 1 1]

Predicted Values:
[0 1 0 0 0 0 1 0 0 0 1 0 1 0 0 0 0 0 0 0 0 1 1 0 0 0 1 1 1 1 0 1 1 0 0 0
0 1 0 0 0 1 0 0 1 0 1 0 0 1 1 0 0 0 1 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 1 1 0
0 0 1 0 1 1 1 1 0 0 0 0 1 1 1 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 1 0
0 1 1]

Variables Terminal

16:34 Python 3

Search

31°C Partly sunny

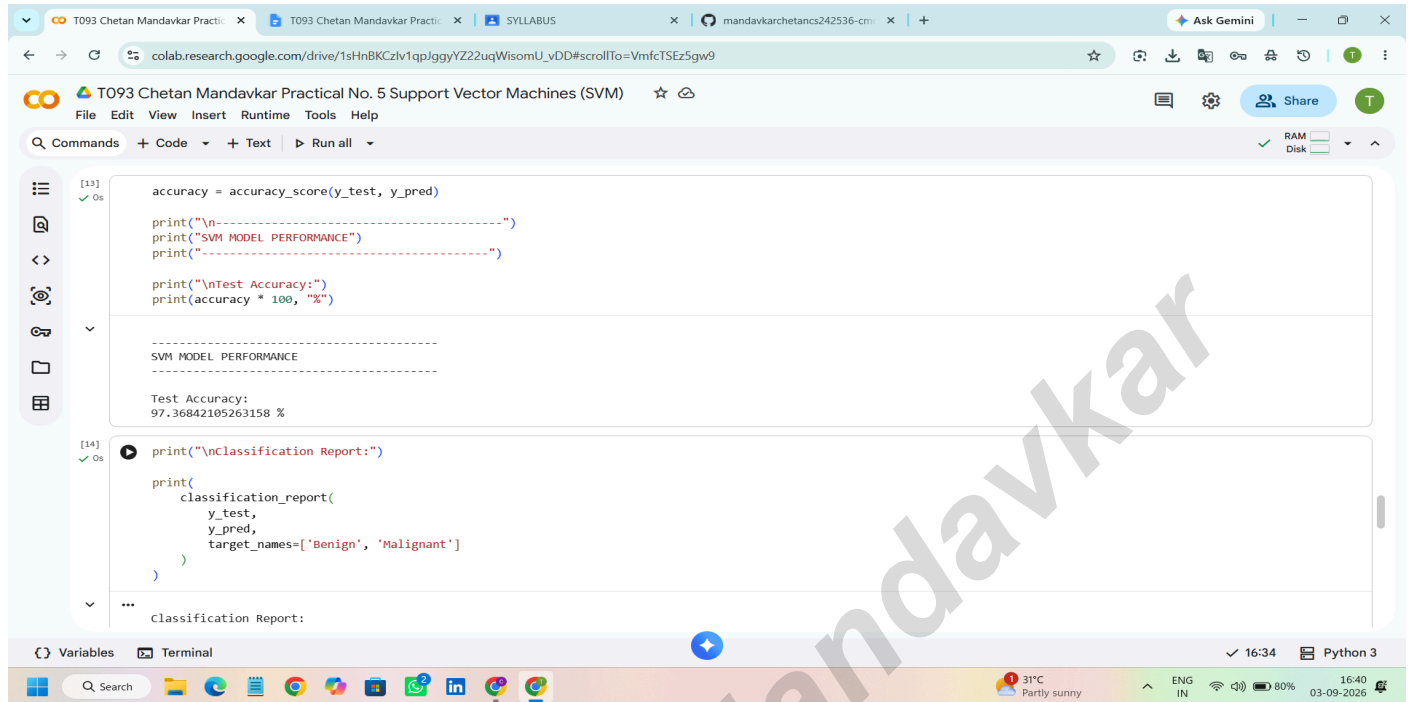
ENG IN

80%

16:40 03-09-2026

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The screenshot shows a Google Colab environment with the title "T093 Chetan Mandavkar Practical No. 5 Support Vector Machines (SVM)". The code cell [13] contains the following Python code:

```
accuracy = accuracy_score(y_test, y_pred)

print("\n-----")
print("SVM MODEL PERFORMANCE")
print("-----")

print("\nTest Accuracy:")
print(accuracy * 100, "%")
```

The output of the code cell [13] is:

```
-----
SVM MODEL PERFORMANCE
-----
Test Accuracy:
97.36842105263158 %
```

The code cell [14] contains the following Python code:

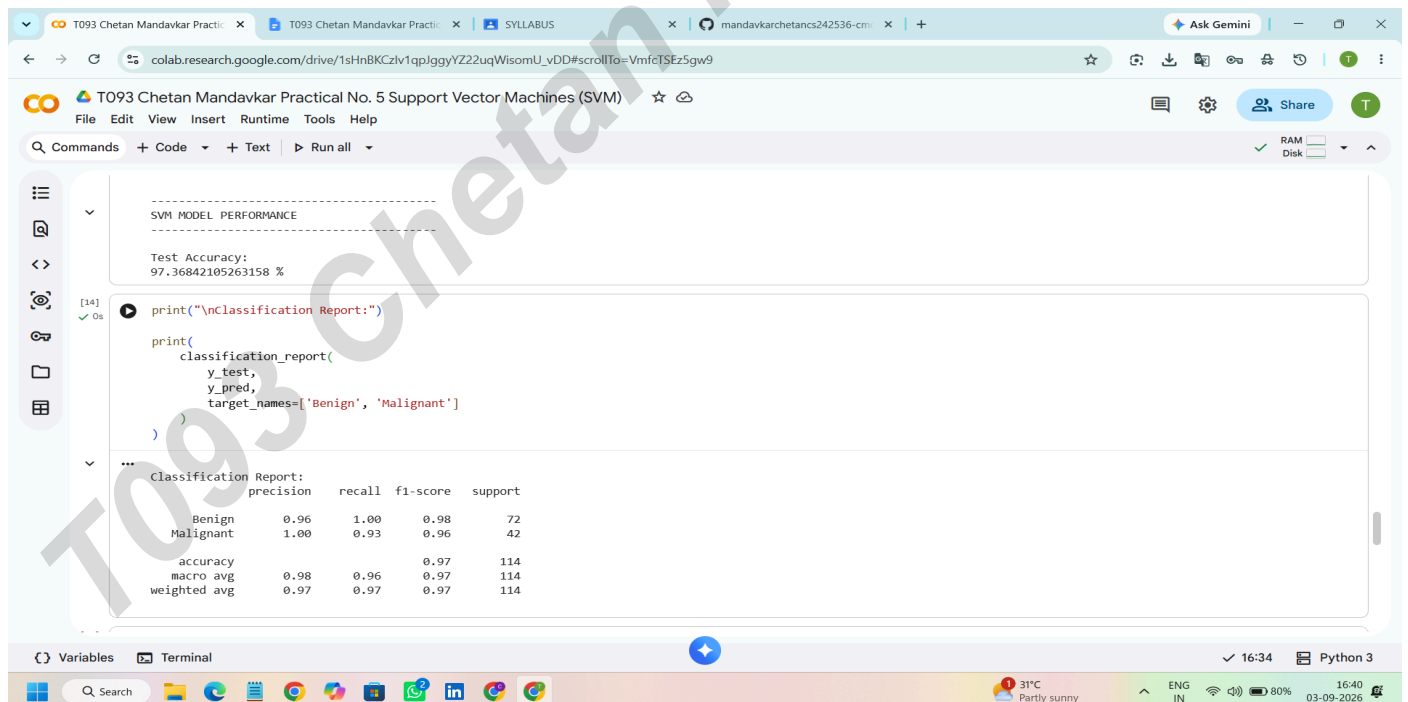
```
print("\nClassification Report:")

print(
    classification_report(
        y_test,
        y_pred,
        target_names=['Benign', 'Malignant']
    )
)
```

The output of the code cell [14] is:

```
Classification Report:
```

The bottom status bar shows the time as 16:34, the language as Python 3, and the date as 03-09-2026.



The screenshot shows the same Google Colab environment as the previous one. The code cell [14] is the same as in the previous screenshot. The output of the code cell [14] is:

```
Classification Report:
precision    recall  f1-score   support

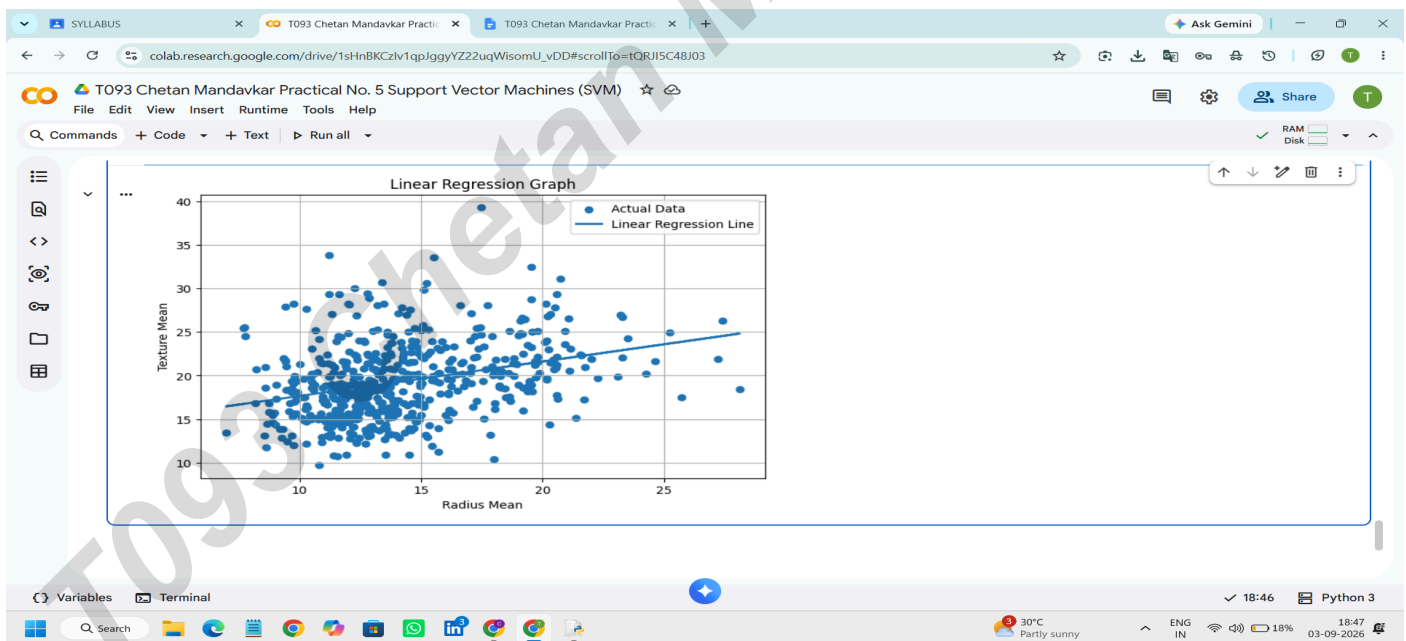
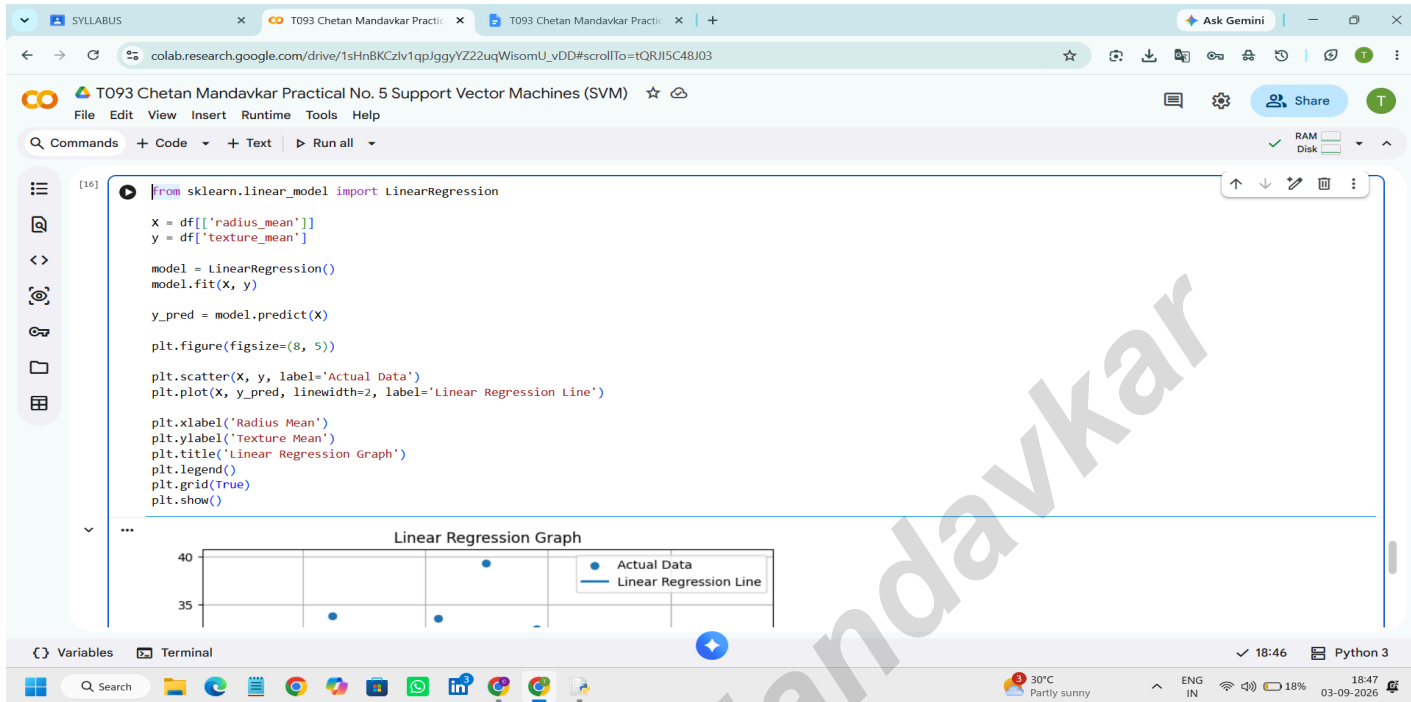
   Benign    0.96    1.00    0.98        72
  Malignant    1.00    0.93    0.96        42

 accuracy    0.98    0.96    0.97       114
  macro avg   0.98    0.96    0.97       114
 weighted avg   0.97    0.97    0.97       114
```

The bottom status bar shows the time as 16:34, the language as Python 3, and the date as 03-09-2026.

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